

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2012

CL103.01 Mechanics of Solids

Date: 14.05.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain principle of transmissibility. [03]
(b) The resultant of two forces P and Q is at right angles to P. Show that the angle between the forces is $\cos^{-1}(-P/Q)$. [04]
- Q - 2 (a) Define scalar and vector quantities. [02]
(b) Two rollers of same diameter are supported by a vertical wall and an inclined plane having inclination 15° with horizontal as shown in **Figure 1**. The upper and lower rollers are respectively 200N and 250N in weight. Assuming smooth surfaces, find the reactions induced at the point of support A, B, C and D. [05]

OR

- (b) A plate measuring 6m x 4m is acted upon by a set of forces in its plane as depicted in **Figure 2**. Determine the magnitude, direction and position of the resultant force with respect to point A. [05]
(c) Locate centroid of an area shown in **Figure 3**. [07]

OR

- (c) Determine the coordinates of the for the lamina shown in **Figure 4**. The shaded area is the opening in the lamina. [07]
- Q - 3 (a) Determine the forces in all members of the truss shown in **Figure 5**. [07]
(b) Two blocks A and B weighing 50 kg and 80 kg respectively are in equilibrium in the position shown in **Figure 6**. Calculate the force P required to move the lower block B and tension in the cable. Take coefficient of friction at all contact surfaces to be 0.3. [07]

OR

- Q - 3 (a) A truss has been loaded and supported as shown in **Figure 7**. Determine the forces in all members of the truss. [07]
(b) A uniform ladder of length 5m and weight 20 N is placed against a smooth vertical wall with its lower end 4m away from the wall. If the ladder is just to slip, determine:
1. The coefficient of friction between the ladder and floor
2. The frictional force acting on the ladder at the point of contact between ladder and floor. [07]

SECTION - II

- Q - 4 (a) Define: Hardness, Ductility, Load factor, Factor of Safety and Proof Stress [05]
(b) Explain elasticity. [02]

Q - 5 Attempt any two:

- Derive relation between Young's Modulus and Shear Modulus with usual notations.
- Under what axial tensile load, the diameter of a steel bar will be reduced from 8cm to 7.995cm? Take E as $2 \times 10^3 \text{ kN/cm}^2$ and Poisson's ratio as 0.3.
- Determine the elongation of cylindrical bar shown in Figure 8. Assume E for the bar material as 200 kN/m^2 .

[14]

Q - 6 Attempt any two:

- Draw shear force and bending moment diagrams for the beam shown in Figure 9.
- Draw shear force and bending moment diagrams for the beam shown in Figure 10.
- Draw shear force and bending moment diagrams for the beam shown in Figure 11.

[14]

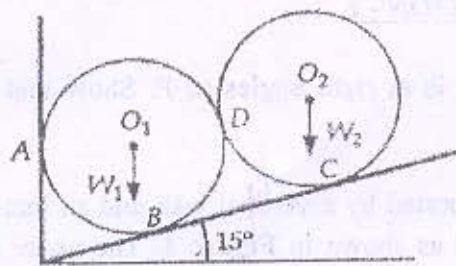


Figure 1

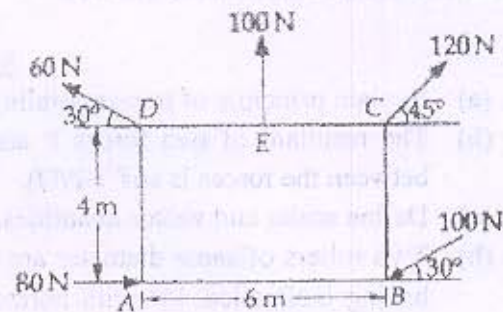


Figure 2

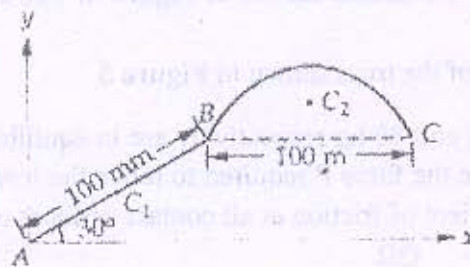


Figure 3

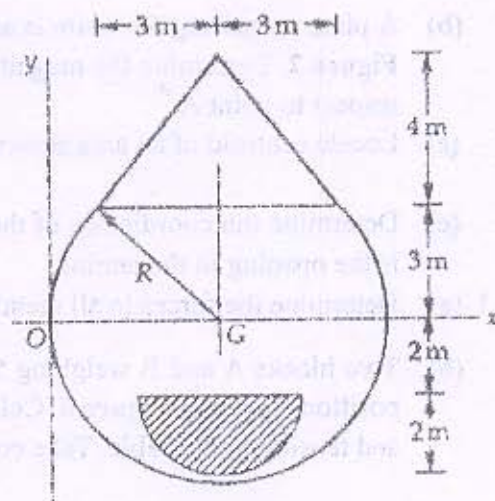


Figure 4

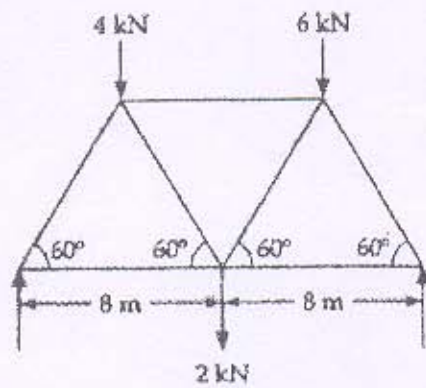


Figure 5

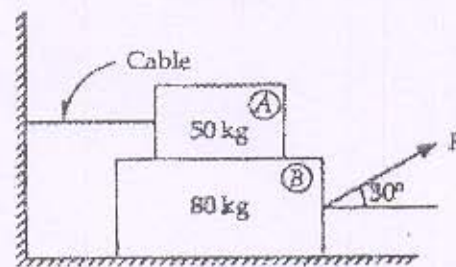


Figure 6

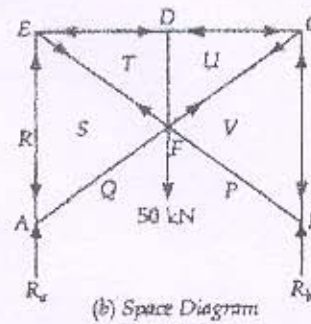
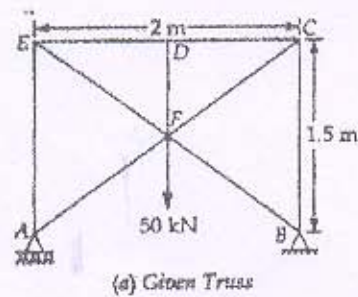


Figure 7

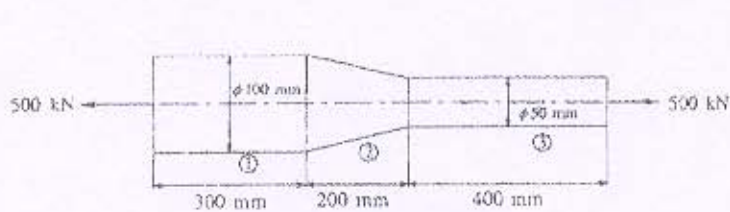


Figure 8

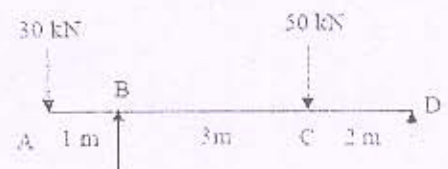


Figure 9

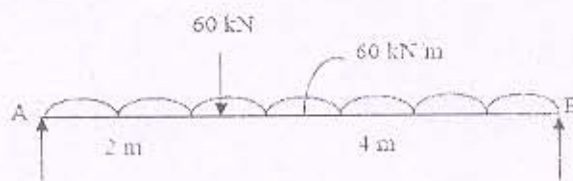


Figure 10

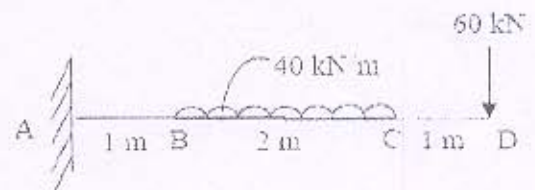


Figure 11
